

An exact polynomial and rational certification of the symmetric five-node Lebesgue optimum

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For quartic Lagrange interpolation at $-1, -a, 0, a, 1$, we give a self-contained certification of the unique minimizing parameter and the sharp Lebesgue constant. Their decimal values, and a degree-73 integer relation for an associated optimum, have already appeared in the literature. We print a complete primitive degree-73 polynomial for the parameter, prove its irreducibility over $\mathbb{Q}$, and isolate the relevant root and sharp value in rational intervals of widths 10^{-18} and $3 \cdot 10^{-18}$. The global argument reduces the Lebesgue function to two quartic peaks whose heights move strictly in opposite directions. Exact rational, Sturm, Sylvester-determinant and finite-field computations are supplied in a replayable supplement. No global priority claim is made.

1 Statement and relation to earlier work

For $0 < a < 1$, put $T_a = \{-1, -a, 0, a, 1\}$ and define

$$\ell_t^{(a)}(x) = \prod_{s \in T_a \setminus \{t\}} \frac{x-s}{t-s}, \quad L_a(x) = \sum_{t \in T_a} |\ell_t^{(a)}(x)|, \quad \Lambda(a) = \max_{-1 \leq x \leq 1} L_a(x).$$

Every terminating decimal below denotes its exact rational value.

Theorem 1. *The set of global minimizing parameters of Λ on $(0, 1)$ is the singleton $\{a_*\}$. Let $P \in \mathbb{Z}[X]$ be the polynomial printed in Appendix A. It is primitive and irreducible over $\mathbb{Q}$, has degree 73, and a_* is its unique real root in*

$$I = [0.620911304689912040, 0.620911304689912041]. \quad (1)$$

In particular, $[\mathbb{Q}(a_) : \mathbb{Q}] = 73$, and the sharp value satisfies*

$$\Lambda(a_*) \in [1.559490209875555263, 1.559490209875555266]. \quad (2)$$

The decimal optimizer $a = 0.620911304689912\dots$ and decimal sharp constant $\Lambda = 1.5594902098\dots$ are already in the literature. Rack and Vajda [2], Remark 7.5, give the five-node, degree-four sharp constant and state that it satisfies an integer polynomial of degree 73. Dumont-Le Brazidec and Peter [3] give the symmetric pair ± 0.620911304689912 added to $\{0, -1, 1\}$. Hesthaven [4], Table 1, lists the degree-four optimal constant as 1.5595. Thus neither the numerical optimizer, the numerical constant, nor the bare existence of a degree-73 relation is claimed here as a first discovery. The degree assertion in Theorem 1 concerns the parameter a_* ; it does not assert an algebraic degree for the sharp value.

The general canonical-node uniqueness and equioscillation characterization are classical results of de Boor and Pinkus [1]. In the present symmetric family the four gap maxima are (M_1, M_0, M_0, M_1) , in the notation below. Their equality therefore instantiates the classical

equal-peak condition. The affine change $z = (x + 1)/2$ preserves every cardinal ratio and the Lebesgue norm, so the formulation on $[0, 1]$ has nodes $0, (1 - a)/2, 1/2, (1 + a)/2, 1$.

The contribution presented here is the complete printed polynomial for the parameter, its irreducibility and algebraic-degree proof, the global branch argument, and explicit rational certificates that can be replayed. Our literature comparison is bounded and does not establish priority for this combination. The historical comparison uses the supplied inspection records for the four cited works; none is a mathematical dependency of the proof. Adjacent computational context includes polynomial evaluation and interpolation in Zeng [5], and exact certificates for sharp bounds in Zeng [6]. Those two works are cited only for these broader subjects, not as results about this Lebesgue optimum.

2 Two quartics and their unique peaks

Reflection gives $\ell_t^{(a)}(-x) = \ell_{-t}^{(a)}(x)$, so L_a is even. In node order $(-1, -a, 0, a, 1)$ the cardinal signs are $(+, -, +, +, -)$ on $(0, a)$ and $(-, +, -, +, +)$ on $(a, 1)$. Indeed the denominator signs are $(+, -, +, -, +)$, and each numerator has constant sign on a gap. Adding the signed cardinals yields

$$F(a, x) = 1 + \frac{x(a - x)(1 + x)(a^2 + ax - a - x + 1)}{a^2(1 - a)}, \quad 0 \leq x \leq a, \quad (3)$$

$$G(a, x) = 1 + \frac{(x^2 - a^2)(1 - x)((2 - a^2)x + 2(1 - a^2))}{a^2(1 - a^2)}, \quad a \leq x \leq 1. \quad (4)$$

Consequently, with

$$M_0(a) = \max_{0 \leq x \leq a} F(a, x), \quad M_1(a) = \max_{a \leq x \leq 1} G(a, x),$$

we have $\Lambda(a) = \max\{M_0(a), M_1(a)\}$.

Lemma 1. *For every $a \in (0, 1)$, each of the two gaps has exactly one stationary point of L_a , and that point is its strict maximum.*

Proof. Set $r = 1 + a^2/(1 - a) > 1$ and $s = 2(1 - a^2)/(2 - a^2) > 0$. Then

$$\begin{aligned} F(a, x) - 1 &= \frac{x(a - x)(1 + x)(r - x)}{a^2}, \\ G(a, x) - 1 &= \frac{(2 - a^2)(x - a)(x + a)(1 - x)(x + s)}{a^2(1 - a^2)}. \end{aligned}$$

Each expression is positive inside its gap and zero at the endpoints. The second logarithmic derivatives are, respectively,

$$\begin{aligned} -\frac{1}{x^2} - \frac{1}{(a - x)^2} - \frac{1}{(1 + x)^2} - \frac{1}{(r - x)^2} &< 0, \\ -\frac{1}{(x - a)^2} - \frac{1}{(x + a)^2} - \frac{1}{(1 - x)^2} - \frac{1}{(x + s)^2} &< 0. \end{aligned}$$

In both cases the first logarithmic derivative decreases continuously from $+\infty$ to $-\infty$ across the gap. Its unique zero is the unique stationary point, with positive derivative before it and negative derivative after it. The value there exceeds the endpoint value 1. $\square$

Write the two maximizing locations as $u(a)$ and $v(a)$. For use in the exact certificate, direct differentiation gives

$$F_x(a, x) = \frac{p(a, x)}{a^2(1-a)}, \quad G_x(a, x) = \frac{q(a, x)}{a^2(1-a^2)},$$

where

$$p(a, x) = 4(1-a)x^3 - 3ax^2 + (2a^3 - 2a^2 + 2a - 2)x + a^3 - a^2 + a, \quad (5)$$

$$q(a, x) = (4a^2 - 8)x^3 + 3a^2x^2 + (4 - 2a^4)x - a^4. \quad (6)$$

3 Opposite motion of the peak heights

Transfer both gaps to $[0, 1]$ by setting

$$H_0(a, y) = F(a, ay), \quad H_1(a, y) = G(a, a + (1-a)y).$$

On every compact parameter subinterval of $(0, 1)$ these functions are uniformly continuous. Since the difference of two maxima is bounded in absolute value by the uniform difference of the functions, both M_i are continuous on $(0, 1)$.

Exact differentiation yields

$$\partial_a H_0(a, y) = \frac{ay(1-y)(1+y)[2-a-2(1-a)^2y]}{(1-a)^2} > 0 \quad (0 < a, y < 1), \quad (7)$$

because the bracket is at least $a(3-2a) > 0$. For the other gap, the polynomial identity

$$\partial_a H_1(a, y) = \frac{y(1-y)Q(a, y)}{a^3(a+1)^2}, \quad Q(a, y) = \sum_{i=0}^6 \sum_{j=0}^2 b_{ij} \mathcal{B}_i^6(a) \mathcal{B}_j^2(y) \quad (8)$$

uses the Bernstein basis $\mathcal{B}_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}$ and the following complete coefficient table:

i	b_{i0}	b_{i1}	b_{i2}
0	0	-2	-8
1	-2/3	-10/3	-10
2	-28/15	-16/3	-12
3	-4	-317/40	-273/20
4	-36/5	-21/2	-71/5
5	-10	-11	-12
6	-4	-4	-4

Identity (8) can be checked by differentiation and expansion. Every Bernstein basis factor is positive on the open square; all coefficients are nonpositive and all but one are negative. Thus $Q < 0$ there and $\partial_a H_1 < 0$.

If $a < b$, choose an interior maximizer of $H_0(a, \cdot)$. Evaluating $H_0(b, \cdot)$ at that point proves $M_0(b) > M_0(a)$. Choosing an interior maximizer of $H_1(b, \cdot)$ instead proves $M_1(a) > M_1(b)$. Therefore

$$D(a) = M_0(a) - M_1(a) \quad \text{is continuous and strictly increasing on } (0, 1). \quad (9)$$

This argument does not differentiate the maximizing locations.

4 Rational certification of the crossing

Let a_L and a_R denote the endpoints of I in (1). The exact brackets for the stationary points are as follows; each right endpoint is the listed left endpoint plus 10^{-30} .

Parameter	Root	Left endpoint
a_L	$u(a_L)$	0.318621732752274017212554831256
a_L	$v(a_L)$	0.833712434791180387794077605697
a_R	$u(a_R)$	0.318621732752274017811626835192
a_R	$v(a_R)$	0.833712434791180388165988567715

At both parameters, substitution into (5)–(6) gives a positive sign at the left endpoint of each bracket and a negative sign at its right endpoint. Lemma 1 identifies these roots with the actual cell maxima. Exact rational Horner interval evaluation of (3)–(4) yields the following peak enclosures. Each right endpoint is the listed left endpoint plus 10^{-26} .

Parameter	Peak	Left endpoint
a_L	$M_0(a_L)$	1.55949020987555526342911289
a_L	$M_1(a_L)$	1.55949020987555526731564134
a_R	$M_0(a_R)$	1.55949020987555526544070371
a_R	$M_1(a_R)$	1.55949020987555526363504357

In particular, direct interval subtraction, with outward rounding, gives

$$\begin{aligned} -3.88652846 \cdot 10^{-18} &\leq D(a_L) \leq -3.88652845 \cdot 10^{-18}, \\ 1.80566013 \cdot 10^{-18} &\leq D(a_R) \leq 1.80566014 \cdot 10^{-18}. \end{aligned}$$

For clarity, the tighter difference bounds use the unrounded rational Horner intervals before the decimal rounding displayed in the table. The supplement reconstructs the cardinals over $\mathbb{Q}$ and checks the peak bounds, all four bracket signs, and the crossing signs. Its Sturm sequences also count exactly one derivative root in each full cell and exactly one in each bracket. Floating-point root finding is unnecessary.

By (9) there is exactly one crossing $a_* \in (a_L, a_R)$. If $a < a_*$, then $\Lambda(a) = M_1(a)$, which strictly decreases; if $a > a_*$, then $\Lambda(a) = M_0(a)$, which strictly increases. Hence a_* is the unique global minimizer on the entire interval $(0, 1)$. The crossing also gives

$$M_0(a_L) \leq M_0(a_*) = \Lambda(a_*) \leq M_0(a_R),$$

and the table proves (2). No assumption at the excluded parameters 0 and 1, and no unexamined stationary branch, is involved.

5 Elimination, irreducibility and root isolation

Put $h_* = \Lambda(a_*) - 1$ and define

$$\begin{aligned} f(a, x, h) &= x(a - x)(1 + x)(a^2 + ax - a - x + 1) - ha^2(1 - a), \\ g(a, x, h) &= (x^2 - a^2)(1 - x)((2 - a^2)x + 2(1 - a^2)) - ha^2(1 - a^2). \end{aligned}$$

The inner and outer stationary peaks give common roots of (f, f_x) and (g, g_x) at (a_*, h_*) . Define $A, B \in \mathbb{Z}[a, h]$ by the exact quotients

$$\text{Res}_x(f, f_x) = a^2(a - 1)A(a, h), \tag{10}$$

$$\text{Res}_x(g, g_x) = a^2(a - 1)(a + 1)(a^2 - 2)B(a, h). \tag{11}$$

These identities, including polynomial divisibility, follow by expanding the Sylvester determinants. Both A and B are cubic in h , with a -degrees 14 and 18. Their complete coefficient arrays are included in the supplement. A second elimination gives

$$\text{Res}_h(A, B) = 256a^8(a-1)^6(a+1)P(a), \quad (12)$$

where P is printed in Appendix A. Every factor removed in (10)–(12) is nonzero for $0 < a < 1$. Thus $A(a_*, h_*) = B(a_*, h_*) = 0$ and $P(a_*) = 0$. Only the necessary direction of elimination is used; no other root of P is asserted to describe an admissible optimizer.

The supplement verifies the three identities using integer Sylvester matrices and fraction-free Bareiss determinants. For the first two identities the respective bidegree bounds in (a, h) are $(21, 3)$ and $(28, 3)$: the determinant has three quartic rows and four derivative rows, with coefficient a -degrees at most 3 or 4, and only the three quartic rows depend affinely on h . The right sides respect the same bounds. Equality is checked at

$$(a, h) \in \{2, \dots, 23\} \times \{0, 1, 2, 3\}, \quad (a, h) \in \{2, \dots, 30\} \times \{0, 1, 2, 3\},$$

respectively. The last resultant has a -degree at most $3 \cdot 14 + 3 \cdot 18 = 96$, whereas its displayed right side has degree 88; equality is checked at $a = 2, \dots, 98$. A polynomial of degree at most d is determined by $d + 1$ distinct values. Applying this statement in each variable proves the identities. These finite grids certify polynomial identities, not sampled optimization assertions.

The algebraic degree

The coefficient gcd of P is 1. Its degree is 73 and its leading coefficient 671088640 is nonzero modulo the prime 229. For the coefficientwise reduction $\bar{P} \in \mathbb{F}_{229}[X]$, exact modular polynomial arithmetic verifies

$$\gcd(\bar{P}, X^{229} - X) = 1, \quad X^{229^{73}} \equiv X \pmod{\bar{P}}. \quad (13)$$

The second congruence is computed by 73 successive modular Frobenius powers starting from X ; the final ascending coefficient list is $(0, 1)$. The first gcd is normalized to the constant 1. Both primalities are checked by trial division.

The polynomial $X^{229^{73}} - X$ is squarefree, and its irreducible factors over $\mathbb{F}_{229}$ have degrees dividing 73. Since 73 is prime, those degrees are 1 and 73. The gcd in (13) excludes degree 1. As $\bar{P}$ itself has degree 73, it is irreducible. Degree preservation and Gauss's lemma imply that P is irreducible over $\mathbb{Q}$. Because $P(a_*) = 0$, the degree of a_* is 73; its monic minimal polynomial over $\mathbb{Q}$ is $P/671088640$.

Isolation of the parameter root

For $l = a_L$, $r = a_R$, expand exactly over $\mathbb{Q}$:

$$P(l + (r - l)t) = \sum_{j=0}^{73} c_j t^j = \sum_{k=0}^{73} \beta_k \mathcal{B}_k^{73}(t), \quad \beta_k = \sum_{j=0}^k c_j \frac{\binom{k}{j}}{\binom{73}{j}}.$$

The rational calculation gives $\beta_0 < 0 < \beta_{73}$ and $73(\beta_{k+1} - \beta_k) > 0$ for all $0 \leq k < 73$. These 73 differences are the Bernstein coefficients of the derivative in t , so the polynomial is strictly increasing on $[0, 1]$. It therefore has exactly one root in I . The global crossing and elimination argument identify that root with a_* . This completes the proof of Theorem 1.

6 Reproducibility, assistance and scope

The supplementary directory contains three verifiers and their exact input data. From that directory run, using Python 3 without optimization:

```
python3 evidence/verify_optimum_certificate.py
python3 evidence/pass3_verify_resultants.py
python3 evidence/pass3_verify_algebra.py
```

All three use only the Python standard library. They check, respectively, the cardinal identities, Sturm counts and rational bounds; the three resultant identities; and irreducibility with rational root isolation. The data include the full polynomial coefficients, the sparse A, B arrays, and all endpoint brackets. The scripts recompute the assertions rather than trusting stored verification flags. The analytic arguments above supply the continuum quantifiers beyond these finite checks.

AI assistance was used in proof development, symbolic computation, verification-code preparation, literature comparison and manuscript drafting. Symbolic discovery used SymPy, and separate implementations recomputed the decisive finite certificates using exact integer and rational arithmetic. An additional model-context mathematical review reconstructed the cardinals, resultants, interval bounds and algebraic degree with a separate implementation. Such review is not external peer review or a formal proof-assistant verification. No floating-point experiment is used as a proof step.

The theorem concerns precisely the symmetric family T_a . It makes no assertion about optimization over arbitrary nonsymmetric node configurations, expression by radicals, or the algebraic degree of $\Lambda(a_*)$. The numerical data and the previously stated degree-73 relation are acknowledged above. The bounded literature comparison does not prove that this explicit polynomial or certificate has never appeared elsewhere.

References

- [1] C. de Boor and A. Pinkus, Proof of the conjectures of Bernstein and Erdős concerning the optimal nodes for polynomial interpolation, *Journal of Approximation Theory* **24** (1978), 289–303. Author-hosted text.
- [2] Rack and Vajda, *Optimal cubic Lagrange interpolation* (2015), Remark 7.5. Journal-hosted text.
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A The complete parameter polynomial

The following table defines $P(X) = \sum_{k=0}^{73} p_k X^k$. Each coefficient, including its sign, is printed in full; the two column pairs together cover every degree from 73 to 0 exactly once.

k	p_k	k	p_k
73	671088640	36	2590136464371211319901
72	-8724152320	35	12984875638311143946527
71	12180258816	34	-7422488026634703911071
70	336349626368	33	-11668409981025051863956
69	-1628262367232	32	11460707061237070479356
68	-3550520279040	31	7806337754198137654608
67	41955928834048	30	-13112302270210059062560
66	-34393961791488	29	-2847870147791332681360
65	-524595019800576	28	11840032707136675318320
64	1353559590264832	27	-1335497890005994756368
63	3358389534257152	26	-8454747373324685067824
62	-17603109112082432	25	3481603554633295820736
61	-3680574819633664	24	4577132067826579016768
60	135052716044764672	23	-3540199448033540127616
59	-135387726825982464	22	-1628964572063109826688
58	-665095342161389312	21	2423229636264373919744
57	1434405543072017344	20	119398331493379373056
56	1928698276451496576	19	-1189760304344953196544
55	-8364439891163274816	18	293559416868863094784
54	-837875944266738448	17	402566863154115100672
53	33214562244938438528	16	-226157769155376791552
52	-23018321220520653896	15	-75841553238739714048
51	-93373209035752053934	14	93162757408012566528
50	137094494847900071050	13	-3714928283791130624
49	175793511373926511751	12	-23482388044527697920
48	-479033750414051075643	11	7151255414183559168
47	-146420919575054257809	10	3106706844186312704
46	1201267022701607448013	9	-2142841358182776832
45	-345189597569286403132	8	22078301572956160
44	-2272720924558662940108	7	302651334035767296
43	1803769259439983079806	6	-74135493175934976
42	3231556717936354876102	5	-14369669397872640
41	-4535501954391456072886	4	9386420707786752
40	-3185705695266529167558	3	-978969075646464
39	8166538514020299577430	2	-279009665482752
38	1280965806327277367790	1	81406810128384
37	-11494378930592932123109	0	-6262062317568